

PRODUCT DATASHEET

PN5321 MINI NFC Module

Compact 13.56 MHz NFC controller module with two included ferrite-backed external antennas

INPUT POWER

3.3-5.0 V DC

HOST LOGIC

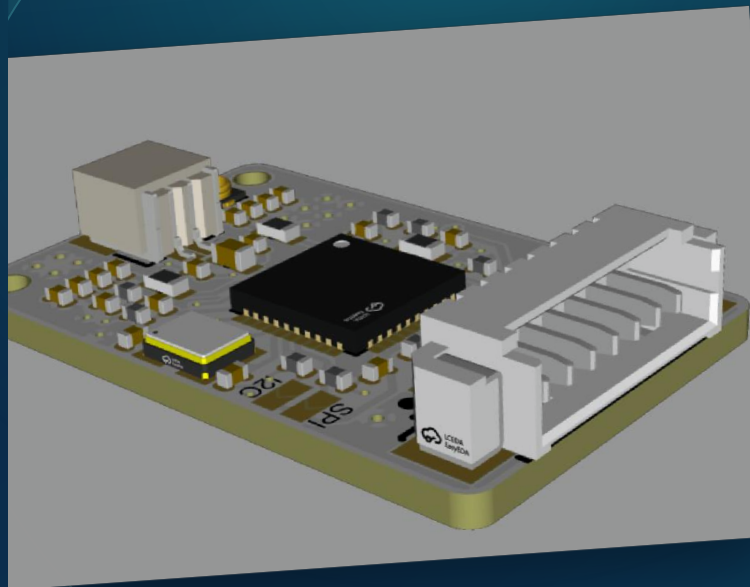
3.3 V

HOST LINKS

UART / I2C / SPI

BOARD SIZE

25 x 16.4 mm



COMPACT NFC INTEGRATION

Small board. Flexible antenna placement.

The PN5321 MINI combines the NXP PN532 NFC controller, selectable UART/I2C/SPI host interfaces and external antenna connectivity on a 25 x 16.4 mm board. Two ferrite-backed antennas are included for compact and extended read-range installations.

13.56 MHz

NFC / HF RFID operating frequency

3 links

UART, I2C and SPI selected by solder pads

2 antennas

10 x 25 mm and 40 x 50 mm, both included

5.8 cm

Stable S50 read distance with 40 x 50 mm antenna

Key advantages

- Compact main board for embedded products and restricted enclosures.
- Ferrite-backed antennas improve installation tolerance near conductive structures.
- One 7-pin host connector carries power, reset and all multiplexed host signals.
- MX1.0 antenna connector for the included wired antennas plus an alternative IPEX4 connector.
- Hardware reset / hard power-down control available at the host connector.
- Optional 0402 I2C pull-up footprints support host-specific bus loading.

Typical applications

A

Access control

Readers, locks, cabinets and attendance terminals.

I

IoT devices

Provisioning, identification and maintenance interfaces.

C

Consumer products

Compact NFC interaction in handheld and desktop products.

O

OEM integration

Custom antenna tuning and production support.

Contents

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PRODUCT SPECIFICATION

Electrical and functional summary

Unless otherwise stated, values in this table describe the complete PN5321 MINI module rather than the PN532 IC alone.

Module		Connections and antennas	
Controller	NXP PN532 / PN5321 ordering family	Host connector	7-pin MX1.25
Operating frequency	13.56 MHz	Wired antenna connector	2-pin MX1.0
Input supply	3.3-5.0 V DC; operation verified at both endpoints	Alternative RF connector	IPEX4
Host I/O level	3.3 V logic	Included antenna 1	10 x 25 mm, ferrite-backed, no adhesive
Host interfaces	HSU (UART), I2C, SPI	Included antenna 2	40 x 50 mm, ferrite-backed, 3M adhesive backing
Default interface	HSU (UART)	Antenna lead	10 cm, MX1.0 plug
Board dimensions	25 x 16.4 x 4.4 mm	Antenna adhesive	Small: none; large: 3M backing
Operating temperature	-25 to +85 °C	Simultaneous antenna use	Not permitted; use one connector only
Compliance	CE, RoHS		

Supported RF technologies

Technology	Supported function	Air data rates
ISO/IEC 14443A / MIFARE	Reader/writer; Type A card emulation	106, 212, 424 kbit/s
ISO/IEC 14443B	Reader/writer	106, 212, 424 kbit/s
FeliCa / JIS X 6319-4	Reader/writer; card emulation	212, 424 kbit/s
ISO/IEC 18092 / NFCIP-1	Initiator and target, active or passive communication	106, 212, 424 kbit/s

! **Protocol limitation:** ISO/IEC 15693 is not supported by the PN532 and is not claimed for this module.

Host interface reference limits

HSU / UART

115200 bit/s, 8-N-1

Default firmware configuration

I2C slave

Up to 400 kHz

7-bit address 0x24; 8-bit write/read 0x48/0x49

SPI slave

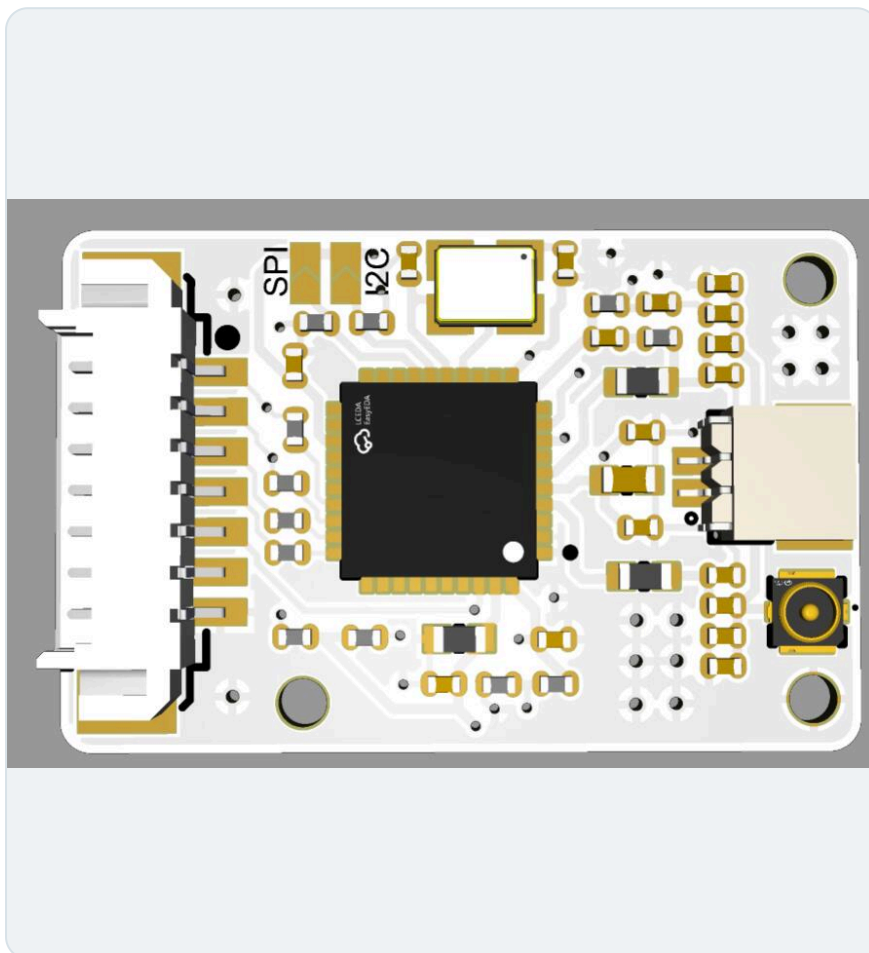
Mode 0, up to 5 MHz

LSB first

BOARD IDENTIFICATION

Hardware overview

The host connector and the two antenna connectors serve different purposes. Do not confuse the 7-pin MX1.25 host connector with the 2-pin MX1.0 antenna connector.



Top-side 3D view. Component appearance may vary by production lot.

1 7-pin MX1.25
Host power, reset and
UART/I2C/SPI signals.

2 2-pin MX1.0
Connector for either included
wired antenna.

3 IPEX4
Alternative RF connector. Never
use together with MX1.0.

4 Mode pads
UART by default; bridge only
the pad marked I2C or SPI.

5 I2C pull-up pads
Two optional 0402 positions,
not populated in production.

6 PN532 controller
13.56 MHz NFC controller with
embedded firmware.

Power

Apply regulated 3.3-5.0 V to the host connector. All host signal levels are 3.3 V.

Antenna

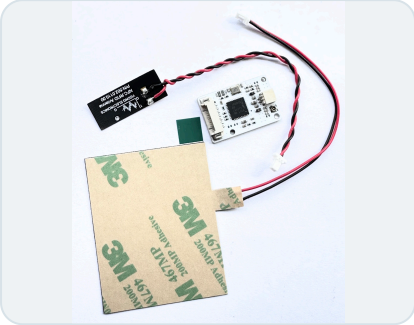
Connect exactly one antenna before enabling RF communication.

Mode

Set solder pads before applying power. Interface selection is sampled during start-up.

7-PIN MX1.25 INTERFACE

Host connector pinout

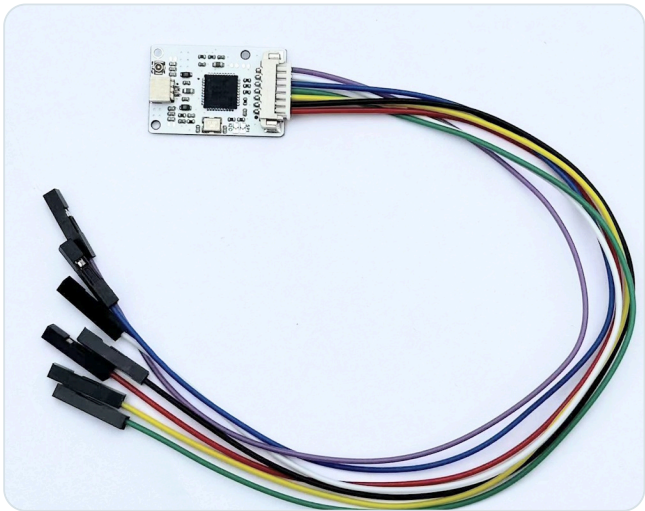


The signal names printed on the PCB are multiplexed according to the selected host interface. Signal direction is stated relative to the module. The product includes the main board and both external antennas shown here.

	Pin	Silkscreen	Direction	Function
7	7	GND	Power	Ground reference.
6	6	5V	Input	Regulated 3.3-5.0 V module supply; pin label remains 5V.
5	5	RST	Input	Active-low hardware reset / hard power-down control. Connected to PN532 RSTPD_N.
4	4	SCK	Input	SPI serial clock. Unused in UART and I2C modes.
3	3	MISO	Output	SPI master-in/slave-out. Unused in UART and I2C modes.
2	2	TX / MOSI / SDA	Multiplexed	UART TX output / SPI MOSI input / I2C SDA open-drain.
1	1	RX / SS / SCL	Multiplexed	UART RX input / SPI NSS input / I2C SCL.

Electrical use

Supply input	3.3-5.0 V DC
Signal high level	3.3 V logic domain
Connector family	MX1.25, 7 positions
Series signal resistors	100 ohm on host interface lines



! 3.3 V logic only: A 5 V microcontroller must not drive module signal pins directly unless a suitable level shifter is used. The pin marked 5V accepts 3.3-5.0 V power and is not the signal voltage.

SOLDER-PAD CONFIGURATION

Select one host interface

The production default is UART. To select another interface, bridge only the solder pad carrying the matching PCB label. Make the change while the module is unpowered.

Default condition	I2C pad	SPI pad	Selected interface	Result
Both pads open	Open	Open	HSU / UART	Factory configuration
-	Bridged	Open	I2C	I2C slave mode
-	Open	Bridged	SPI	SPI slave mode
-	Bridged	Bridged	Reserved	Module does not operate

I2C pad open



Leave open for UART or SPI.

SPI pad open



Leave open for UART or I2C.

I2C pad bridged



Selects I2C when SPI remains open.

SPI pad bridged



Selects SPI when I2C remains open.

1. Remove power

Disconnect module power and all host signals.



2. Set one mode

Bridge only the pad marked I2C or SPI, or leave both open for UART.



3. Reconnect

Apply 3.3-5.0 V and initialize the matching software interface.



Do not bridge I2C and SPI at the same time. This selects the reserved state and the module will not work until one bridge is removed.

REFERENCE WIRING

UART, I2C and SPI connections

The examples below show logical connections. MCU pin names vary by platform. Always connect grounds and keep all host signals in the 3.3 V logic domain.

UART	default mode	I2C	bridge I2C pad	SPI	bridge SPI pad
MCU TX	— RX / SS / SCL	MCU SCL	— RX / SS / SCL	MCU SCK	— SCK
MCU RX	— TX / MOSI / SDA	MCU SDA	— TX / MOSI / SDA	MCU MISO	— MISO
GND	— GND	GND	— GND	MCU MOSI	— TX / MOSI / SDA
3.3-5 V source	— 5V	3.3-5 V source	— 5V	MCU CS	— RX / SS / SCL
Default HSU: 115200 bit/s, 8 data bits, no parity, 1 stop bit.		Up to 400 kHz. 7-bit address 0x24. Install optional pull-ups only when the host does not provide them.		SPI slave, Mode 0, LSB first, SCK up to 5 MHz. Keep NSS high when idle.	

Power and reset common to all modes

Module pin	Connect to	Requirement
5V	Regulated 3.3-5.0 V supply	Use short power and ground wiring with adequate current capability for RF operation. Read-distance data use 5 V; module-input power profiles are identified as 5 V or 3.3 V.
GND	Host ground	A common reference is required for all interfaces.
RST	3.3 V MCU open-drain GPIO, optional	Pull low for hardware reset / hard power-down; release to high impedance for normal operation.

Software

Use an ELECHOUSE PN532 library, libnfc or another PN532-compatible host implementation. Configure the library for the same hardware interface selected on the module.

Start-up check

Before RF testing, read the PN532 firmware version or execute a basic diagnostic command to verify the host link and frame protocol.

! **Interface exclusivity:** only one host interface can be active. Do not connect two host buses to the multiplexed pins simultaneously.

CONTROL AND OPTIONAL ASSEMBLY

Reset input and I2C pull-up pads

The host RST pin is connected to PN532 pin 38, RSTPD_N. It is an input to the module, not the RSTOUT_N status output.

RST behavior

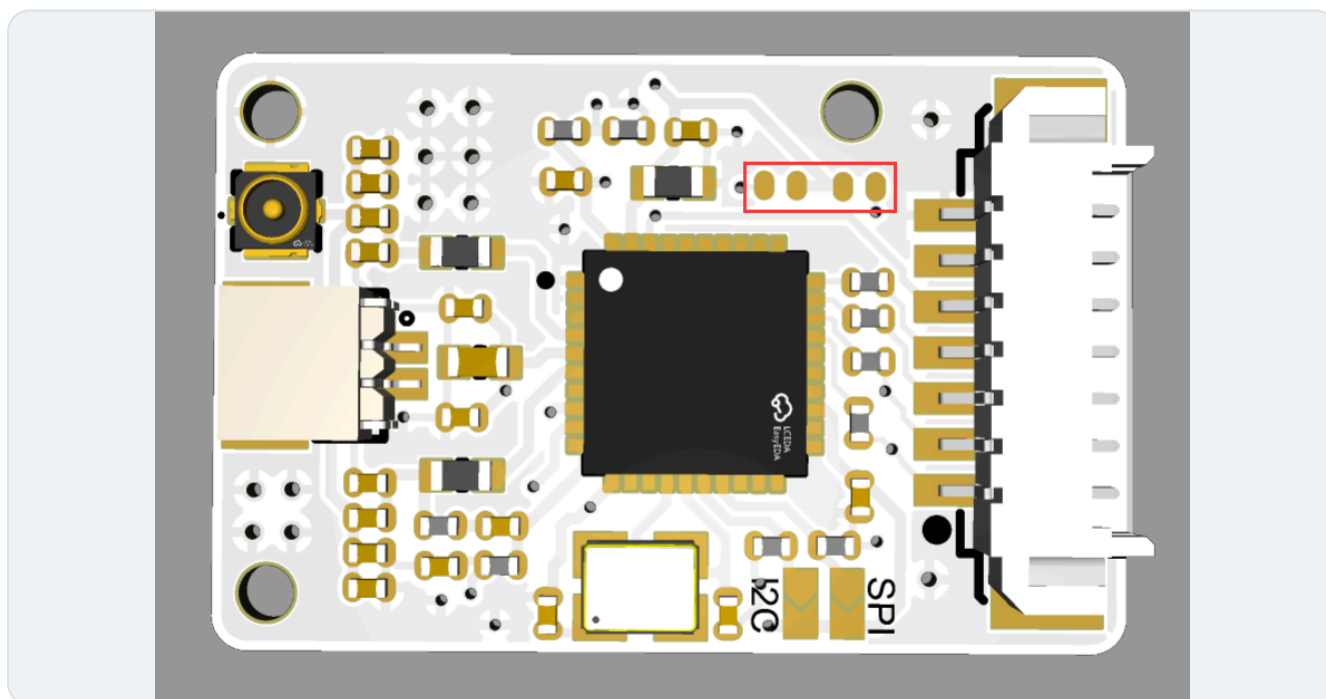
Low	PN532 enters reset and Hard Power-Down; oscillator and internal current sources are disabled.
Released / high	Module bias releases RSTPD_N and the PN532 performs its power-up sequence.
Recommended drive	3.3 V open-drain output: drive low to reset, switch to high impedance to release.

i RSTOUT_N, PN532 pin 26, is not routed to the host connector on this hardware version.

I2C pull-up assembly

Two 0402 footprints connect SDA and SCL to 3.3 V. Production boards leave both positions unpopulated.

Default	DNP / NC
Recommended	4.7 kohm, 0402, 1%
High capacitance at 400 kHz	2.2 to 3.3 kohm after waveform verification
Host already has pull-ups	Leave module footprints unpopulated



Red box: the two unpopulated 0402 I2C pull-up positions. Both pull up to the module 3.3 V rail.

NOMINAL 5 V MODULE INPUT

5 V power consumption by antenna

Measured results for the two included antennas with a MIFARE S50, 85 x 54 mm PVC white card fixed at the reading position.

Operating state	40 x 50 mm antenna			10 x 25 mm antenna		
	Mean current	100 Hz max	Mean power	Mean current	100 Hz max	Mean power
Ready idle RF field off	19.254 mA	19.259 mA	95.294 mW	16.527 mA	16.531 mA	81.830 mW
Normal 4 Hz card polling	29.812 mA	146.215 mA	145.621 mW	25.056 mA	89.523 mA	123.090 mW
Maximum sustained Continuous RF and polling	143.698 mA	146.685 mA	684.266 mW	90.278 mA	91.884 mA	435.884 mW
PowerDown	0.188 mA	0.192 mA	0.942 mW	0.188 mA	0.192 mA	0.941 mW

Antenna comparison

- During normal polling, the 10 x 25 mm antenna reduced mean current by 15.95%.
- During maximum sustained operation, it reduced mean current by 37.18%.
- PowerDown current was approximately 0.19 mA with either antenna.

Supply planning

- The 40 x 50 mm antenna produced the higher tested load and should be used for conservative 5 V supply sizing.
- Its maximum sustained input averaged 4.762 V and reached a measured minimum of 4.754 V.
- Select the antenna using both power and read-range performance in the final enclosure.

i Test condition: HSU/UART operation with the card centered over the antenna. Values characterize the tested module and antenna assemblies.

NOMINAL 3.3 V MODULE INPUT

3.3 V power consumption by antenna

Measured results for the two included antennas with a MIFARE S70, 55 x 85 mm PVC white card fixed at the reading position.

Operating state	40 x 50 mm antenna			10 x 25 mm antenna		
	Mean current	100 Hz max	Mean power	Mean current	100 Hz max	Mean power
Ready idle RF field off	17.897 mA	17.901 mA	59.251 mW	17.897 mA	17.901 mA	59.267 mW
Normal 4 Hz card polling	25.773 mA	112.143 mA	84.434 mW	23.461 mA	84.631 mA	77.217 mW
Maximum sustained Continuous RF and polling	109.608 mA	111.487 mA	349.706 mW	84.401 mA	85.850 mA	272.499 mW
PowerDown	0.372 mA	0.376 mA	1.241 mW	0.372 mA	0.376 mA	1.241 mW

Antenna comparison

- During normal polling, the 10 x 25 mm antenna reduced mean current by 8.97%.
- During maximum sustained operation, it reduced mean current by 23.00%.
- PowerDown current was approximately 0.372 mA with either antenna.

3.3 V rail margin

- The 40 x 50 mm maximum sustained input averaged 3.1905 V and reached 3.186 V.
- The 10 x 25 mm maximum sustained input averaged 3.2286 V and reached 3.225 V.
- Use a low-impedance 3.3 V source and short power wiring.

i Test condition: HSU/UART operation with the card centered over the antenna. Values characterize the tested module and antenna assemblies.

MEASURED OPERATING PROFILES

Voltage, current and power waveforms

The four profiles show ready idle, normal 4 Hz card polling, continuous RF and polling, and PowerDown.

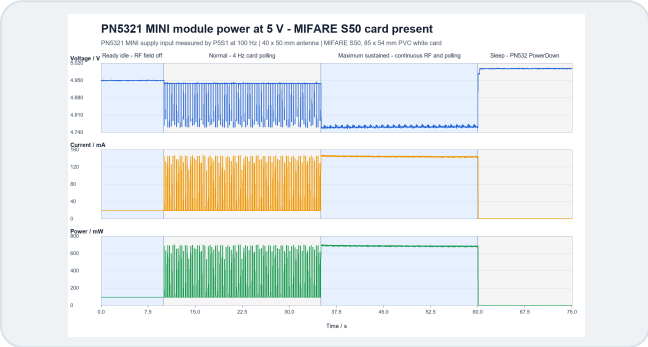


Figure 1. 5 V, 40 x 50 mm antenna, MIFARE S50.

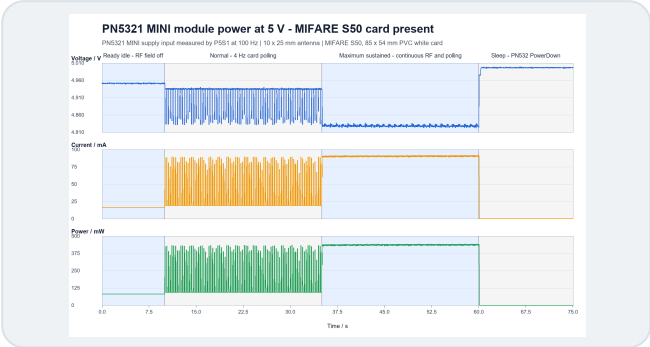


Figure 2. 5 V, 10 x 25 mm antenna, MIFARE S50.

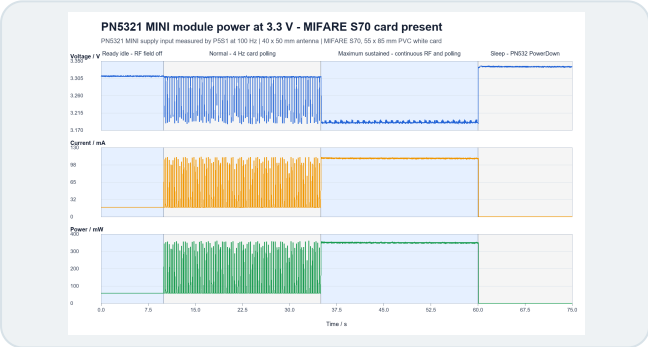


Figure 3. 3.3 V, 40 x 50 mm antenna, MIFARE S70.

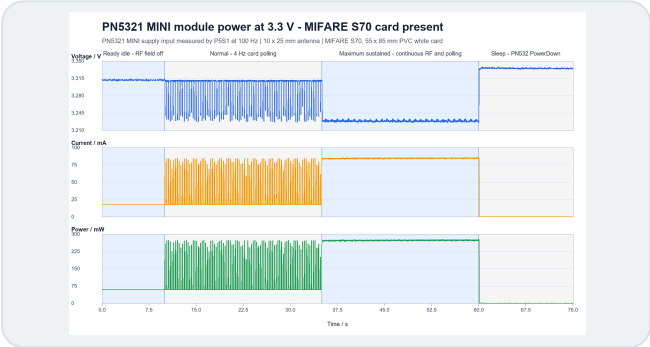


Figure 4. 3.3 V, 10 x 25 mm antenna, MIFARE S70.

Unified test conditions

Parameter	5 V profiles	3.3 V profiles
Credential	MIFARE S50, 85 x 54 mm PVC white card	MIFARE S70, 55 x 85 mm PVC white card
Antennas	40 x 50 mm and 10 x 25 mm	40 x 50 mm and 10 x 25 mm
Normal operation	Card fixed at the reading position; 4 Hz polling	
Maximum sustained	Continuous RF field and continuous polling	
PowerDown	PN532 PowerDown with RF disabled	

! Sampling limitation: maximum values are observations at 100 Hz and do not bound transients shorter than approximately 10 ms. "Maximum sustained" means the highest tested continuous condition, not an absolute electrical rating.

INCLUDED FERRITE-BACKED ANTENNAS

Choose the antenna for the enclosure

Each product includes both antenna sizes. Both use a 10 cm lead, an MX1.0 plug and ferrite backing. The 10 x 25 mm antenna has no adhesive; the 40 x 50 mm antenna has 3M adhesive backing.

**10 x 25 mm**

Best for restricted space and discreet placement.

No adhesive. Custom tuning is available to optimize this antenna for the final installation.

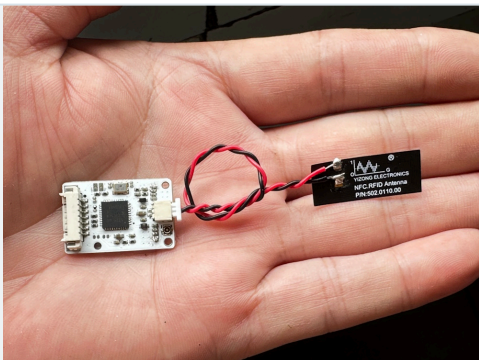
**40 x 50 mm**

Larger magnetic coupling area for increased read distance.

3M adhesive backing. Recommended when the enclosure allows a larger antenna window.

Connector separation

Connector	Purpose	Use
7-pin MX1.25	Host wiring	Power, RST and UART/I2C/SPI. This is not an antenna connector.
2-pin MX1.0	Included wired antennas	Connect either the 10 x 25 mm or 40 x 50 mm antenna.
IPEX4	Alternative RF connection	Use only with a compatible and correctly tuned antenna assembly.



Installation guidelines

- Use only one antenna connector at a time.
- Keep the antenna plane parallel to the expected card plane.
- Keep the antenna centered behind the user interaction area.
- Avoid sharp lead bends and noisy switching-current loops.
- Validate tuning and range in the final enclosure.

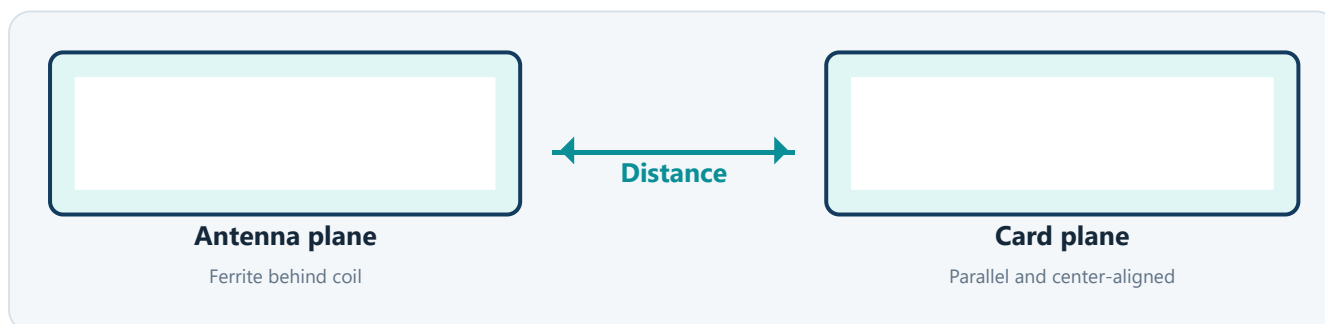
MEASURED MODULE PERFORMANCE

Stable continuous read distance

Measurements were taken at 5 V and 3.3 V module input with the card and antenna parallel, center-aligned and stationary. The values below represent stable, continuous reading rather than an intermittent maximum.

Antenna	Input	MIFARE S50 white PVC card 85 x 54 mm	MIFARE S50 round PVC card 25 mm diameter
40 x 50 mm Ferrite-backed	5 V	5.8 cm	3.3 cm
	3.3 V	5.5 cm	4.2 cm
10 x 25 mm Ferrite-backed	5 V	3.0 cm	2.0 cm
	3.3 V	3.2 cm	2.5 cm

Test geometry



Factors that reduce range

- Offset or angular misalignment between antenna and card.
- Metal, batteries, displays or ground planes near the antenna.
- RF noise from switching converters, displays or motor wiring.
- Different card coil geometry, chip sensitivity or material stack-up.

Small-antenna optimization

ELECHOUSE can tune the 10 x 25 mm antenna for a customer enclosure, surrounding materials and target card. This service can improve practical read performance when the generic tuning is shifted by the final product.

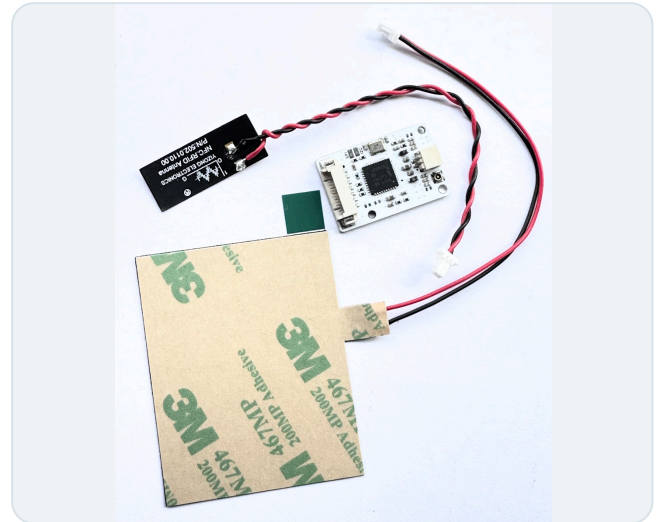
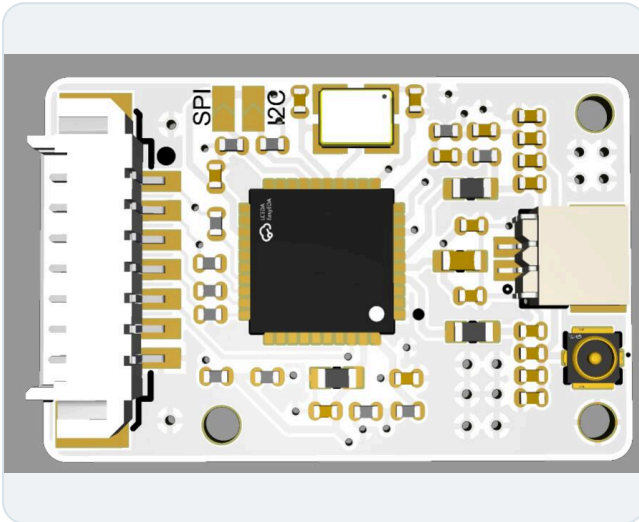
Custom performance must be validated with the production enclosure and target credential.

i Interpreting the results: The table is a controlled comparison of the two included antennas. It is not a guaranteed distance for every credential or installation.

INTEGRATION RESOURCES

Mechanical data and package contents

Use the downloadable CAD and 3D files for detailed placement. Dimensions in this datasheet identify the overall module envelope.

**25 mm**

Board length

16.4 mm

Board width

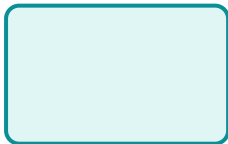
4.4 mm

Maximum stated assembly height

Downloads

CAD / DXP drawing package[MINIV2_dimension.zip](#)**3D model / STEP package**[3D_PN532_MINIV2.zip](#)

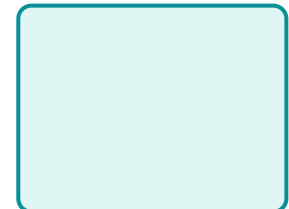
Standard package contents

**PN5321 MINI main board**

Quantity 1

**10 x 25 mm antenna**

Quantity 1, ferrite-backed, no adhesive, 10 cm lead, MX1.0

**40 x 50 mm antenna**

Quantity 1, ferrite-backed, 3M adhesive backing, 10 cm lead, MX1.0

RELEASE INFORMATION

Handling, references and revision history

Handling checklist

- Observe ESD precautions when handling the exposed PCB and RF connectors.
- Apply regulated 3.3-5.0 V only to the supply pin; keep host logic at 3.3 V.
- Set mode solder pads only while the module is unpowered.
- Never bridge the I2C and SPI mode pads simultaneously.
- Connect only one antenna at a time: MX1.0 or IPEX4.
- Do not pull or sharply crease the antenna lead or ferrite layer.
- Verify read performance after final enclosure assembly.

Product identification

Product	PN5321 MINI NFC Module
SKU	NFC_EX_MINI
Hardware	V2
Datasheet	Rev. 1.5
Release	September 2026

Revision history

Revision	Date	Description
1.5	2026-09	Added 3.3 V stable read-distance results for both included antennas and both test cards.
1.4	2026-09	Streamlined the power chapter to 5 V and 3.3 V dual-antenna summaries plus a unified waveform page.
1.3	2026-09	Added 3.3 V profiles for both antennas and corrected the module-input power scope.

References

1. [ELECHOUSE PN5321 MINI product page](#)
2. [NXP PN532/C1 Near Field Communication Controller - Product Data Sheet](#)
3. [NXP UM0701 PN532 User Manual](#)

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